

# $\pi$ -nucleon scattering

$\pi^\pm$  ---   $H_2$  liquid

$\pi^+$  beam against target of p, n

$\pi^+ + p \rightarrow \pi^+ + p$  elastic scattering (a)

$\pi^+ + n \rightarrow \pi^+ + n$  (d)

$\pi^+ + n \rightarrow \pi^0 + p$  (g)

$\pi^- + p \rightarrow \pi^- + p$  (c)

$\pi^- + p \rightarrow \pi^0 + n$  (j)

$\pi^- + n \rightarrow \pi^- + n$  (f)

for principle:

$\pi^0 + n \rightarrow \pi^0 + n$

$\pi^0 + p \rightarrow \pi^+ + n$

$\pi^0 + p \rightarrow \pi^0 + p$

$\tau_{\pi^0} \ll \tau_{\pi^\pm}$

$\pi^0 \rightarrow \gamma\gamma$

we cannot accelerate  $\pi^0$

$\sigma \propto |M_{fi}|^2 \rho(E)$   
 $\rho(E)$  phase space. basically the same for all channels

# events  $\propto \sigma \propto |M_{fi}|^2$

$$M_{fi} = -i \int d^3r \psi_f^\dagger H_I \psi_i$$

$$\psi = \psi_{\text{space}} \psi_{\text{spin}} \psi_{\text{isospin}}$$

(f)  $\pi^- + n \rightarrow \pi^- + n$

$$|\pi^- \rangle = |I=1, I_3=-1\rangle$$

$$|n \rangle = |1/2, -1/2\rangle$$

$$|1/2, -1/2\rangle |1/2, 1/2\rangle = |3/2, -1/2\rangle$$

$$\pi^+ p \rightarrow \pi^+ p$$

$$|2, +1\rangle |1/2, +1/2\rangle = |3/2, +3/2\rangle$$

$$(C) \quad \pi^- + p \rightarrow \pi^- + p \quad M_C$$

$$|2, -1\rangle |1/2, +1/2\rangle = \alpha |3/2, -1/2\rangle + \beta |1/2, -1/2\rangle$$

$$I_3 = -1 + 1/2 = -1/2. \quad \text{could be } I = 1/2, 3/2$$

$\alpha, \beta$ : Clebsch-Gordan coeff.

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}} |3/2, -1/2\rangle + \sqrt{\frac{2}{3}} |1/2, -1/2\rangle$$

$$\alpha = \sqrt{\frac{1}{3}} \quad \beta = -\sqrt{\frac{2}{3}}$$

| Pion    | Nucleon | $I = \frac{3}{2}$   |                      |                      |                | $I = \frac{1}{2}$     |                       |
|---------|---------|---------------------|----------------------|----------------------|----------------|-----------------------|-----------------------|
|         |         | $I_3 = \frac{3}{2}$ | $\frac{1}{2}$        | $-\frac{1}{2}$       | $-\frac{3}{2}$ | $\frac{1}{2}$         | $-\frac{1}{2}$        |
| $\pi^+$ | $p$     | 1                   |                      |                      |                |                       |                       |
| $\pi^+$ | $n$     |                     | $\sqrt{\frac{1}{3}}$ |                      |                | $\sqrt{\frac{2}{3}}$  |                       |
| $\pi^0$ | $p$     |                     | $\sqrt{\frac{2}{3}}$ |                      |                | $-\sqrt{\frac{1}{3}}$ |                       |
| $\pi^0$ | $n$     |                     |                      | $\sqrt{\frac{2}{3}}$ |                |                       | $\sqrt{\frac{1}{3}}$  |
| $\pi^-$ | $p$     |                     |                      | $\sqrt{\frac{1}{3}}$ |                |                       | $-\sqrt{\frac{2}{3}}$ |
| $\pi^-$ | $n$     |                     |                      |                      | 1              |                       |                       |

$$M_C = (\langle 3/2, -1/2 | \alpha + \langle 1/2, -1/2 | \beta) H_I (\alpha | 3/2, -1/2\rangle + \beta | 1/2, -1/2\rangle)$$

$\nexists$   $H_I$  (strong interaction) conserves Isospin.

$$M_C = \alpha^2 \overbrace{\langle 3/2, -1/2 | H_I | 3/2, -1/2\rangle}^{M_3} + \beta^2 \underbrace{\langle 1/2, -1/2 | H_I | 1/2, -1/2\rangle}_{M_1} \\ = \frac{1}{3} M_3 + \frac{2}{3} M_1 = \frac{1}{3} (M_3 + 2M_1)$$

$$\sigma_C(\pi^- + p \rightarrow \pi^- + p) \propto |M_C|^2 = \frac{1}{9} |M_3 + 2M_1|^2$$

$$(a) \quad \pi^+ + p \rightarrow \pi^+ + p$$

$$|\pi^+ p\rangle = |1, +1\rangle |1/2, +1/2\rangle = |3/2, 3/2\rangle$$

$$M_a = \langle 3/2, 3/2 | H_I | 3/2, 3/2\rangle = M_3$$

$$\sigma_a(\pi^+ + p \rightarrow \pi^+ + p) \propto |M_a|^2 = |M_3|^2$$

$$(j) \quad \pi^- + p \rightarrow \pi^0 + n$$

$$|\pi^- p\rangle = |1, -1\rangle |1/2, 1/2\rangle = \sqrt{\frac{1}{3}} |3/2, -1/2\rangle - \sqrt{\frac{2}{3}} |1/2, -1/2\rangle$$

$$I_3 = -1/2$$

$$|\pi^0 n\rangle = |1, 0\rangle |1/2, -1/2\rangle = \sqrt{\frac{2}{3}} |3/2, -1/2\rangle + \sqrt{\frac{1}{3}} |1/2, -1/2\rangle$$

$$I_3 = -1/2 \quad I_1 = 1/2, 3/2$$

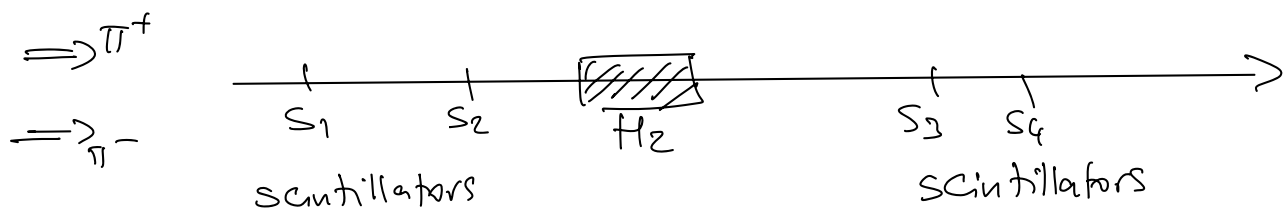
$$M_j = \langle \pi^0 n | H_I | \pi^- p \rangle = \frac{\sqrt{2}}{3} M_3 - \frac{\sqrt{2}}{3} M_1 = \frac{\sqrt{2}}{3} (M_3 - M_1)$$

$$\sigma_j(\pi^- p \rightarrow \pi^0 n) \propto |M_j|^2 \propto \frac{2}{9} |M_3 - M_1|^2$$

$$\sigma_a(\pi^+ p \rightarrow \pi^+ p) \propto |M_a|^2 = |M_3|^2 \rho(\pi p)$$

$$\sigma_j(\pi^- p \rightarrow \pi^0 n) \propto |M_j|^2 \propto \frac{2}{9} |M_3 - M_1|^2 \rho(\pi p)$$

$$\sigma_c(\pi^- p \rightarrow \pi^- p) \propto |M_c|^2 = \frac{1}{9} |M_3 + 2M_1|^2 \rho(\pi p)$$



$S_1$  AND  $S_2$

$\Rightarrow \#$  of  $\pi^\pm$  incoming.

$S_3$  AND  $S_4$

$\Rightarrow \#$   $\pi^\pm$  coming out

Experimentally

$$R = \frac{\#(\pi^+ p \rightarrow \pi^+ p)}{\#(\pi^- p \rightarrow \text{anything})} = \frac{\sigma_a}{\sigma_c + \sigma_j}$$

$$R = \frac{|M_3|^2}{\frac{2}{9}|M_3 - M_1|^2 + \frac{1}{9}|M_3 + 2M_1|^2}$$

$$M_3: \langle I = 3/2 | \pi\pi | I = 3/2 \rangle$$

$$M_1: \langle I = 1/2 | \pi\pi | I = 1/2 \rangle$$

limit 1)  $|M_1| \gg |M_3|$

$$R = \frac{|M_3|^2}{\frac{2}{9}|M_1|^2 + \frac{1}{9}4|M_1|^2} \ll 1$$

$\#(\pi^+ p \rightarrow \pi^+ p) \ll \#(\pi^+ p \rightarrow \text{anything})$   
Not confirmed by data

limit 2)  $|M_1| \ll |M_3|$

$$R = \frac{|M_3|^2}{\frac{2}{9}|M_3|^2 + \frac{1}{9}|M_3|^2} = 3$$

$$\begin{aligned} \underline{P}_1 &= (E_\pi, P_\pi, 0, 0) \xrightarrow{\pi^+} \pi^+ \\ \underline{P}_2 &= (m_p, 0, 0, 0) \xrightarrow{p} p \end{aligned}$$

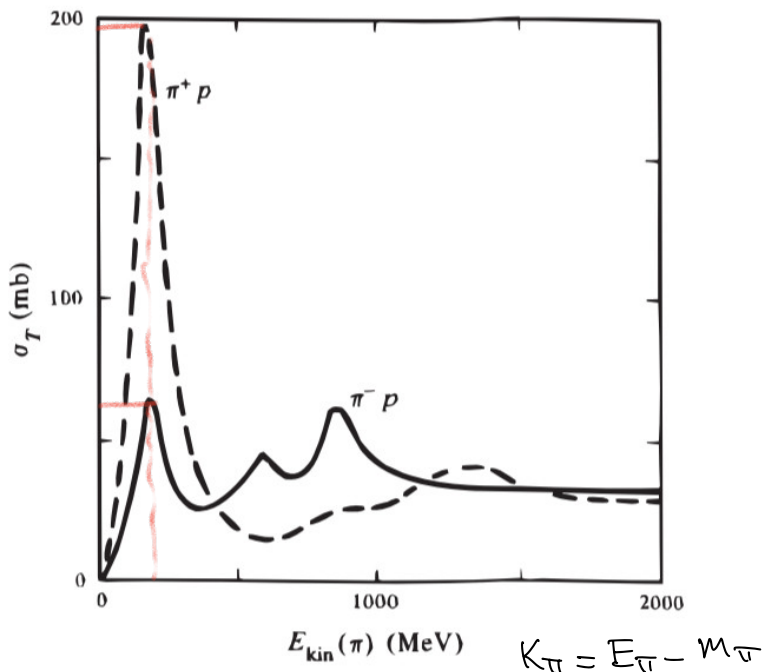


Figure 5.35: Total cross section as a function of pion kinetic energy for the scattering of positive and negative pions from protons. (1 mb = 1 millibarn =  $10^{-27}$  cm<sup>2</sup>.)

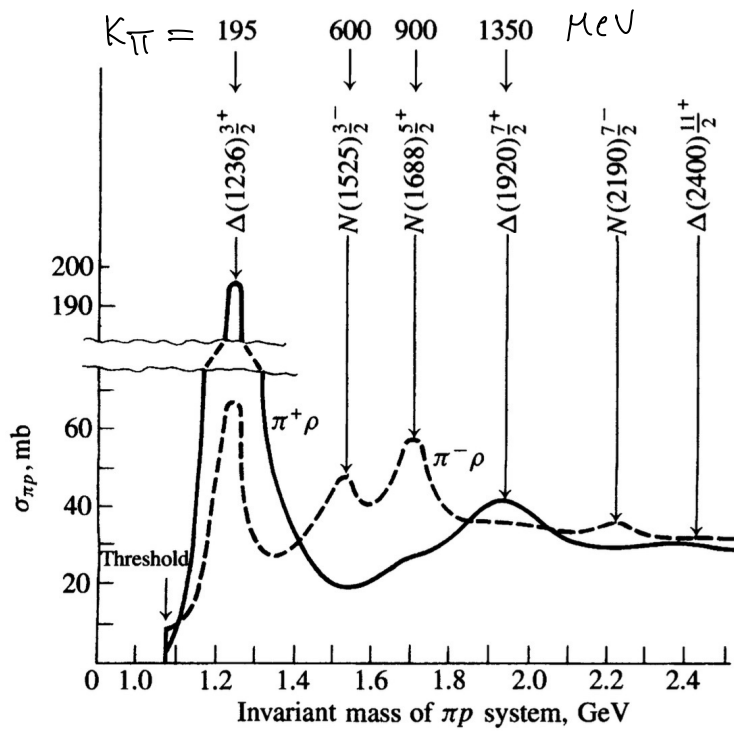
$$\begin{aligned} S &= (\underline{P}_1 + \underline{P}_2)^2 = \\ &= m_\pi^2 + m_p^2 + 2\underline{P}_1 \cdot \underline{P}_2 \\ &= m_\pi^2 + m_p^2 + 2E_\pi \cdot m_p \\ &= m_\pi^2 + m_p^2 + 2m_p(m_\pi + K_\pi) \\ &= (m_\pi + m_p)^2 + 2m_p K_\pi \end{aligned}$$

$$\Rightarrow K_\pi = \frac{S - (m_\pi + m_p)^2}{2m_p}$$

$$K_\pi = 0.2 \text{ GeV} \Rightarrow \sqrt{S} = 1.2 \text{ GeV}$$

$$\pi^+ p \rightarrow \Delta^{++}(1232) \rightarrow \pi^+ p$$

hadronic  
resonance



$$\Delta^{++} \quad I = 3/2$$

$$N \text{ baryon: } I = 1/2$$

$$\text{for } \sqrt{s} = m_\Delta$$

$$u_3 = \langle 3/2 | H | 3/2 \rangle$$

is dominating.

$$\text{For } \sqrt{s} = m_N.$$

$\Rightarrow M_1$  dominates.

$\Rightarrow$  Experimental proof of isospin as a new symmetry.

Beam of pions:

$$\Delta x_{\text{LAB}} = \beta \gamma c \tau$$

pion mean decay length in LAB.

$$\beta \gamma = \frac{p}{m} = \frac{1.4 \text{ GeV}}{0.14 \text{ GeV}} \approx 10$$

$$\tau = 10^{-8} \Rightarrow \Delta x_{\text{LAB}} = 10 \times 3 \times 10^8 \frac{\text{m}}{\text{s}} \times 10^{-8} = 30 \text{ m}$$

$$e^- + p \rightarrow e^- + X \quad \text{QED} \quad \propto \alpha_{\text{EM}}^2 \quad \alpha_{\text{QCD}} \gg \alpha_{\text{EM}}$$

$$p + p \rightarrow p + p + \pi^+ + \pi^- \quad \text{QCD} \quad \propto \alpha_{\text{QCD}}^2$$

## Discrete Symmetries

Parity, Charge Conjugation.

$$\text{parity } \mathbb{P} \psi(\vec{x}) = a \psi(-\vec{x})$$

Inversion operators

$$\mathbb{P} \vec{p} \rightarrow -\vec{p}$$

parity inversion on external coordinates  $\vec{x}$

$$\mathbb{P} \mathbb{P} \psi(\vec{x}) = a^2 \psi(\vec{x}) \Rightarrow \mathbb{P} = \mathbb{P}^{-1}$$

$$\mathbb{P}^2 = \mathbb{1}$$

what about  $\mathbb{P}^\dagger$

$$\langle x | \mathbb{P} | x' \rangle = \langle x | -x' \rangle = \delta(x+x')$$

$$\langle x | \mathbb{P}^\dagger | x' \rangle = \langle -x | x' \rangle = \delta(-x-x') = \delta(x+x')$$

$$\Rightarrow \mathbb{P}^\dagger = \mathbb{P} = \mathbb{P}^{-1}$$

This group has two elements:  $\mathbb{I}$   $\mathbb{P}$

$\mathbb{P}$  is Hermitian.  $\Rightarrow$  eigenvalues can be physical constants of motion.

$$\mathbb{P}^2 \psi = a^2 \psi = \psi \Rightarrow a = \pm 1$$

parity eigenvalue is multiplicative.

$$\mathbb{P} \psi = \mathbb{P}(\psi_1 \psi_2) = \mathbb{P} \psi_1 \mathbb{P} \psi_2 = a_1 a_2 \psi_1 \psi_2 = a_1 a_2 \psi.$$

$a_{12} = a_1 \cdot a_2$

In QFT intrinsic parity of elementary particles defined by convention

elementary fermions  $\left\{ \begin{array}{l} \text{leptons } l^-, \nu \\ \text{quarks} \end{array} \right\} \quad p = +1$

QFT  $\Rightarrow$  anti-fermions  $\left\{ \begin{array}{l} l^+, \bar{\nu} \\ \bar{q} \end{array} \right\} \quad p = -1$

bosons and anti-bosons have  $p = +1$

Composite states

state with  $p_1 p_2$  (two particles)

$$P_{12} = \underbrace{p_1 p_2}_{\text{intrinsic parity}} (-1)^L$$

$L$ : relative angular momentum.

proton:  $uud$   $L=0$  proton is ground state of  $uud$  combination.

$$P_{\text{prot}} = (+1)^3 (-1)^L \\ = +1$$

neutron:  $udd$   $P_n = (+1)^3 (-1)^0 = +1$

$$\pi^{\pm}: u\bar{d} \quad P_{\pi} = (+1)(-1)(-1)^0 \\ = -1$$

pion is ground state of  $u\bar{d}$

photon  $\gamma$

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In QFT  $P_{\gamma} = -1$

Reminder about action of parity:

$$\vec{p} \quad \underline{P} \vec{p} = -\vec{p} \quad p = -1 \text{ for vectors}$$

scalar:  $s = \vec{p} \cdot \vec{x}$

$$P(s) = (P\vec{p}) \cdot (P\vec{x}) = (-\vec{p}) \cdot (-\vec{x}) = \vec{p} \cdot \vec{x}$$

$$\underline{P} s = s. \quad \text{scalar here } p = +1$$

pseudo-vector / polar-vector / axial-vector

$$\vec{a} = \vec{b} \times \vec{c} \quad \vec{S} \text{ spin or angular mom.}$$

$$P\vec{a} = P\vec{b} \times P\vec{c} = \vec{b} \times \vec{c} = \vec{a} \quad \text{pseudo-vector } p = +1.$$

pseudo-scalar:

$$a = \vec{S} \cdot \vec{p}$$

has parity of  $-1$ .

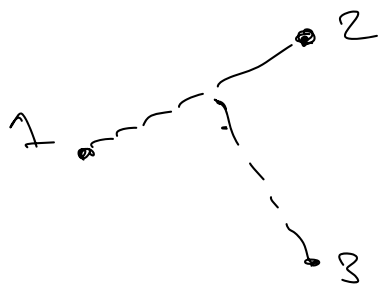
( $\vec{S}$  spin or any pseudo vector)

$$Pa = (P\vec{S}) \cdot (P\vec{p}) = \vec{S} \cdot (-\vec{p}) = -a$$

$\pi^{\pm}, \pi^0$  pseudo-scalar mesons

## system of multiple particles

$$|4\rangle = |\pi^+ \pi^- \pi^0\rangle$$



$$\begin{aligned} & P P \\ & P P \pi^+ \pi^- \\ & P P \pi^0 \pi^0 \pi^+ \end{aligned}$$

$L_{12}$ : consideration of 1-2 system.

$L_3$ : c.m. of 3 with respect to center of mass of 1-2

3 particles:  $P = P_1 \cdot P_2 \cdot P_3 (-1)^{L_{12}} (-1)^{L_3}$

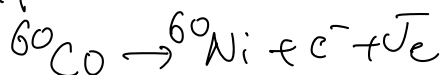
Conservation Law for parity:

EM ✓

Strong ✓

Weak ✗

WU experiment in 1957



## Charge Conjugation

Inversion in intrinsic variable.

$$C\psi = a\psi$$

$$C^2\psi = a^2\psi$$

$$a^2 = +1.$$

$$\Rightarrow a = \pm 1$$

$C$ : particle  $\rightarrow$  anti-particle

all quantum numbers  $\rightarrow$   $(-1)$  quantum number

Exception:  $\vec{S}, \vec{X}, t$

Example of quantum numbers that change:

$Q_e, I_3, \text{Baryon number, lepton flavor number,}$

$C$ : eigenstates? only electrically neutral particles.

Candidates:  $\gamma, n, \pi^0, \nu$

$C\gamma = ?$   $-1$  in EM  $\vec{E}, \vec{I}$  change sign under  $C$



$$\mathbb{C} |\gamma\rangle = -|\gamma\rangle$$

$$\mathbb{C} \gamma \rightarrow \gamma \equiv \bar{\gamma}$$

$\gamma$  is its own anti particle.

neutron

$$n \rightarrow p e^- \bar{\nu}_e$$

$$\bar{n} \rightarrow \bar{p} e^+ \nu_e$$

→ Can be distinguished experim.  
by charge of  $e^+/e^-$

$$\Rightarrow n \neq \bar{n}$$

⊥  
not eigenstate of  $\mathbb{C}$

$$\underline{\pi^0 \equiv \bar{\pi}^0}$$

eigenstate of  $\mathbb{C}$ .

$$\pi^0 \rightarrow \gamma \gamma$$

$$C_{\pi^0} = (C_{\gamma})^2 = +1. \quad \Rightarrow \quad C_{\gamma} = \pm 1.$$

$$\pi^0 \rightarrow \gamma \gamma \gamma$$

$$C = +1 \quad (C_{\gamma})^3 = -1$$

⊗ in EM

not seen so far

$$\underline{U \stackrel{?}{=} \bar{U}}$$

Open question

#### $\pi^0$ DECAY MODES

For decay limits to particles which are not established, see the appropriate Search sections ( $A^0$  (axion) and Other Light Boson ( $X^0$ ) Searches, etc.).

| Mode                                    | Fraction ( $\Gamma_i/\Gamma$ )   | Scale factor/<br>Confidence level |
|-----------------------------------------|----------------------------------|-----------------------------------|
| $\Gamma_1$ $2\gamma$                    | (98.823 ± 0.034) %               | S=1.5                             |
| $\Gamma_2$ $e^+ e^- \gamma$             | (1.174 ± 0.035) %                | S=1.5                             |
| $\Gamma_3$ $\gamma$ positronium         | (1.82 ± 0.29) × 10 <sup>-9</sup> |                                   |
| $\Gamma_4$ $e^+ e^+ e^- e^-$            | (3.34 ± 0.16) × 10 <sup>-5</sup> |                                   |
| $\Gamma_5$ $e^+ e^-$                    | (6.46 ± 0.33) × 10 <sup>-8</sup> |                                   |
| $\Gamma_6$ $4\gamma$                    | < 2 × 10 <sup>-8</sup>           | CL=90%                            |
| $\Gamma_7$ $\nu \bar{\nu}$              | [a] < 2.7 × 10 <sup>-7</sup>     | CL=90%                            |
| $\Gamma_8$ $\nu_e \bar{\nu}_e$          | < 1.7 × 10 <sup>-6</sup>         | CL=90%                            |
| $\Gamma_9$ $\nu_\mu \bar{\nu}_\mu$      | < 1.6 × 10 <sup>-6</sup>         | CL=90%                            |
| $\Gamma_{10}$ $\nu_\tau \bar{\nu}_\tau$ | < 2.1 × 10 <sup>-6</sup>         | CL=90%                            |
| $\Gamma_{11}$ $\gamma \nu \bar{\nu}$    | < 6 × 10 <sup>-4</sup>           | CL=90%                            |

#### Charge conjugation (C) or Lepton Family number (LF) violating modes

|                                       |    |                           |        |
|---------------------------------------|----|---------------------------|--------|
| $\Gamma_{12}$ $3\gamma$               | C  | < 3.1 × 10 <sup>-8</sup>  | CL=90% |
| $\Gamma_{13}$ $\mu^+ e^-$             | LF | < 3.8 × 10 <sup>-10</sup> | CL=90% |
| $\Gamma_{14}$ $\mu^- e^+$             | LF | < 3.4 × 10 <sup>-9</sup>  | CL=90% |
| $\Gamma_{15}$ $\mu^+ e^- + \mu^- e^+$ | LF | < 3.6 × 10 <sup>-10</sup> | CL=90% |